

MAT 426, Class 12 Homework: QR Factorization
(Due Thursday, March 19, 2026)

1. Fill in the missing entries of the matrix Q so make Q an orthogonal matrix.

$$(a) \quad Q = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{3}} & * \\ 0 & \frac{1}{\sqrt{3}} & * \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{3}} & * \end{bmatrix} \qquad (b) \quad Q = \begin{bmatrix} \frac{1}{2} & \frac{2}{\sqrt{14}} & * & * \\ \frac{1}{2} & \frac{1}{\sqrt{14}} & * & * \\ \frac{1}{2} & 0 & * & * \\ \frac{1}{2} & -\frac{3}{\sqrt{14}} & * & * \end{bmatrix}$$

2. The columns of A were obtained by applying the Gram-Schmidt Process to the columns of A . Find an upper triangular matrix R with positive entries on the diagonal such that $A = QR$. State the modified matrix Q if necessary.

$$(a) \quad A = \begin{bmatrix} 2 & 8 & 2 \\ 1 & 7 & -1 \\ -2 & -2 & 1 \end{bmatrix}, \quad Q = \begin{bmatrix} \frac{2}{3} & \frac{1}{3} & \frac{2}{3} \\ \frac{1}{3} & \frac{2}{3} & -\frac{2}{3} \\ -\frac{2}{3} & \frac{2}{3} & \frac{1}{3} \end{bmatrix} \qquad (b) \quad A = \begin{bmatrix} 1 & 3 \\ 2 & 4 \\ -1 & -1 \\ 0 & 1 \end{bmatrix}, \quad Q = \begin{bmatrix} \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} \\ \frac{2}{\sqrt{6}} & 0 \\ -\frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} \\ 0 & \frac{1}{\sqrt{3}} \end{bmatrix}$$

3. Find a QR decomposition for the following matrices.

$$(a) \quad A = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix} \qquad (b) \quad A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & 2 \\ -1 & 1 & 0 \\ 1 & 5 & 1 \end{bmatrix} \qquad (c) \quad A = \begin{bmatrix} 1 & 3 & 5 \\ -1 & -3 & 1 \\ 0 & 2 & 3 \\ 1 & 5 & 2 \\ 1 & 5 & 8 \end{bmatrix}$$

4. Suppose that A is an orthogonal matrix. Find a QR factorization of A .
5. Prove that A is invertible if and only if $A = QR$, where Q is orthogonal and R is upper triangular with nonzero entries on its diagonal.
6. Suppose $A = QR$ where Q is a $m \times n$ and R is $n \times n$. Prove that if the columns of A are linearly independent, then R must be invertible.
7. Finish Example 2 of the Class 12 Handout.